



Vijaybhoomi School of Science & Technology

MID-TERM EXAMINATIONS AUGUST 2026

Course Name: Generative AI

Course Code: STAI345

Duration: 1.00 Hrs

Max. Marks: 20

Instructions to the students:

- 1. Attempt all questions.**
 - 2. Submit your results on the LMS according to the submission requirements, and appear for a viva-voce according to the schedule communicated to you separately.**
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Coding Test / Hackathon, to be followed by a viva voce.

You must design a system that allows a user to query your database using a raw input file (image, audio clip or video frame) and retrieves the "nearest neighbours" from your dataset. Crucially, you must benchmark and compare how different embedding architectures affect the quality of these search results.

You are strictly prohibited from using textual data or metadata for the similarity search. The search must be purely based on the latent vector representation of the content.

- Select a modality: Audio, Image, Video, 3D Mesh, etc.
- Acquire a dataset (minimum 1,000 samples) that is suitable for similarity search.
- Implement pipelines for the embedding model. Try to look out for embedding models that work well with different modalities.
- Generate vector embeddings for your entire dataset using both models. Visualize the difference in how they cluster data (using t-SNE or PCA).
- Ingest them into a vector database of your choice. Look into what VDBs support what modalities best.
- Use appropriate metrics like recall rate and query latency to evaluate performance.
- Present a working demonstration of the model during the viva.

Submission requirements:

- A .zip folder containing all the code files, with appropriate markdowns and documentation.



The rubrics used for evaluation during the viva are as follows:

1. Data Pipeline & Modality Strategy (15%)
2. Embedding Model Experimentation (30%)
3. Vector Database Mechanics & Indexing (30%)
4. Evaluation & Benchmarking (25%)

"Thou shouldst not have been old till thou hadst been wise."

(Fool, King Lear, Act, 1 Scene 5 - William Shakespeare)
